

QAM-DETR: Quality-Aware Mamba DETR for Audio Moment Retrieval from Long Audio

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Abstract—This paper presents QAM-DETR, our system for DCASE 2026 Challenge Task 6: Audio Moment Retrieval from Long Audio. The task requires localizing the temporal segment in a long audio recording that best matches a given natural-language query. Unlike clip-level audio-text retrieval, audio moment retrieval demands precise boundary prediction, long-range temporal modeling, and reliable ranking of candidate moments.

QAM-DETR builds on a DETR-style moment localization framework using frozen pretrained audio and text representations. LAION-CLAP is used as the primary audio-language representation, while MS-CLAP, WavLM, and RoBERTa features are selectively integrated through lightweight learnable gates to exploit complementary acoustic and linguistic information. QAM-DETR progressively integrates these representations, constructs query-conditioned temporal features through 3-way cross-modal interaction and bidirectional temporal modeling, and explicitly predicts localization quality for candidate ranking. For selected final system variants, a frozen audio-language LLM verifier is further used as a post-processing re-ranker, which re-scores candidate segments without modifying their predicted temporal boundaries or updating model parameters.

Experiments on the CASTELLA development splits show that the proposed components consistently improve the DETR-style baseline, with BiMamba temporal encoding and quality-aware ranking providing substantial gains. Final ensemble systems further improve retrieval performance by combining diverse feature configurations and, in selected variants, LLM-based candidate re-ranking. The final system achieved joint first place in DCASE 2026 Task 6, demonstrating the effectiveness of quality-aware temporal modeling and candidate re-ranking for long-audio moment retrieval.

Index Terms—Audio moment retrieval, temporal localization, audio-text cross-attention, BiMamba

1. INTRODUCTION

Audio moment retrieval localizes the start and end of the temporal segment in a long recording that matches a natural-language query. Unlike clip-level audio-text retrieval, it requires semantic matching, long-range temporal reasoning, precise boundaries, and reliable candidate ranking.

DCASE 2026 Challenge Task 6 formalizes this problem using long recordings and natural-language queries [1]. Target events may occupy short intervals, while queries can describe high-level concepts. Evaluation at high temporal Intersection over Union (IoU) thresholds emphasizes boundary accuracy. Pretrained CLAP-style representations provide semantic alignment, but AMR also requires frame-level reasoning. Moreover, DETR-style models generate multiple candidate windows whose classification confidence may not reflect localization quality.

To address these issues, we propose QAM-DETR, a quality-aware DETR-style framework designed around long-audio moment retrieval as a progressive localization pipeline. First, heterogeneous pretrained audio and text representations are projected into a shared space and selectively integrated through learnable gates. A 3-way cross-modal interaction then injects query semantics into temporal audio features and incorporates acoustic context into the query representation.

BiMamba and lightweight multi-scale temporal fusion model the resulting query-conditioned sequence bidirectionally across temporal scales. Finally, a temporal IoU quality branch complements semantic foreground confidence with localization reliability during candidate ranking. For selected final system variants, a frozen audio-language LLM serves only as a post-processing verifier, without modifying temporal boundaries or model parameters.

Our primary technical contribution is a unified quality-aware localization pipeline that jointly couples heterogeneous audio-text representations, query-conditioned bidirectional temporal reasoning, and explicit localization-quality estimation for long-audio moment retrieval. Unlike conventional DETR-style localization, the proposed framework separates semantic relevance from localization reliability and integrates both signals for candidate ranking. We further study frozen audio-language LLM verification as a boundary-preserving post-processing strategy and analyze its configuration-dependent benefits for candidate ranking. Our best system variant ranked joint first in the official DCASE 2026 Task 6 evaluation.

2. RELATED WORK

Audio moment retrieval is closely related to audio-text retrieval, but requires temporal grounding of a natural-language query within a long audio recording. Instead of retrieving or ranking an entire audio clip, the model must predict the start and end times of the query-relevant moment. Recent datasets such as Clotho-Moment and CASTELLA have enabled systematic evaluation of language-based audio moment retrieval under long-audio settings, where target events can be short, sparse, and described by free-form captions [2], [3].

Pretrained audio-language models such as LAION-CLAP and MS-CLAP learn shared audio-text representations through contrastive learning and provide strong semantic matching capability [4], [5]. However, clip-level audio-text representations alone are insufficient for audio moment retrieval, because the task requires temporally resolved reasoning, boundary prediction, and reliable candidate ranking. We therefore use pretrained audio and text features as frozen representations and focus on improving cross-modal fusion, temporal modeling, and localization-aware ranking on top of these features.

DETR-style localization models formulate moment retrieval as a set prediction problem, where multiple candidate windows are decoded and matched to ground-truth moments. However, classification confidence does not always reflect localization quality. To improve temporal modeling and candidate selection, we combine a quality prediction branch with a BiMamba-based bidirectional encoder. This design is motivated by Mamba, which provides efficient selective state-space modeling for long sequences [6], and recent Mamba-based moment retrieval models [7]. In addition, we use a frozen audio-language LLM only as a post-processing verifier for top-ranked candidate windows, keeping the DETR-style model responsible for temporal boundary prediction.

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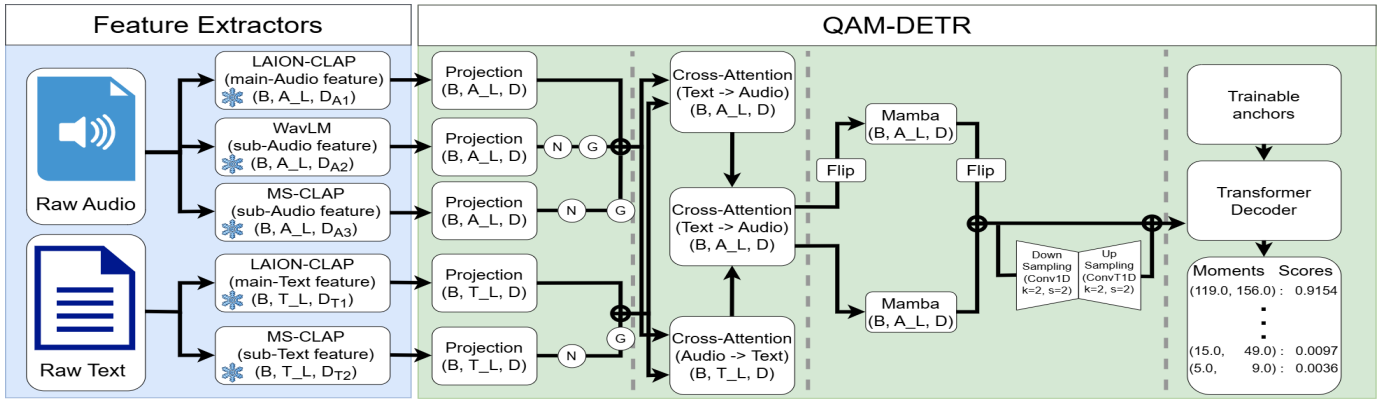


Fig. 1: Overall architecture of QAM-DETR. Pretrained audio and text features are projected, gated, fused through 3-way cross-attention, encoded by BiMamba with multi-scale temporal fusion, and decoded into candidate moments with ranking scores.

3. METHOD

QAM-DETR predicts candidate temporal windows from long audio and a natural-language query, retaining the decoder-based set prediction structure of the DCASE 2026 Task 6 baseline [1]. Fig. 1 connects the three stages detailed below: gated representation integration and query conditioning, bidirectional temporal modeling, and localization-quality-aware ranking.

3.1. Multi-feature cross-modal fusion

To exploit complementary information from multiple pretrained representations, QAM-DETR uses LAION-CLAP as the primary audio-language feature and selectively integrates auxiliary MS-CLAP, WavLM, and RoBERTa features. Each feature stream is projected into a shared hidden dimension using an individual projection layer. Auxiliary streams are then combined with the main LAION-CLAP representation through lightweight learnable scalar gates. Unlike token-wise gating, each auxiliary stream has a single sigmoid-normalized gate shared across all temporal tokens, which keeps the fusion module simple while allowing the model to control the contribution of each representation. The same strategy is applied separately to audio and text streams.

Let x_{main} denote the projected LAION-CLAP feature and x_j denote the j -th projected auxiliary feature. The fused representation is computed as

$$x_{\text{fused}} = x_{\text{main}} + \sum_{j=1}^J \alpha_j x_j, \quad (1)$$

where $\alpha_j = \sigma(g_j)$ is the learnable scalar gate for the j -th auxiliary stream. g_j is a trainable parameter and $\sigma(\cdot)$ denotes the sigmoid function.

Feature-level gating integrates complementary representations within each modality, while temporal localization requires interaction between the fused audio sequence and the query. We therefore apply progressive 3-way cross-modal interaction before temporal encoding, without an additional alignment loss. First, audio features attend to the text query to highlight query-relevant temporal information. Second, text features attend back to the audio sequence to incorporate acoustic context. The final interaction injects the resulting cross-modal information back into the audio sequence. This design produces a query-conditioned temporal audio representation while preserving acoustic context, which is then passed to the temporal encoder for boundary localization.

3.2. BiMamba encoder with multi-scale temporal fusion

BiMamba operates on the query-conditioned temporal representation produced by cross-modal fusion: fusion emphasizes semantics relevant to the query, while bidirectional temporal modeling captures how this information evolves across long audio. This context helps describe events whose interpretation and boundaries depend on surrounding audio. We replace the baseline Transformer encoder with a BiMamba-based temporal encoder, building on selective state-space modeling [6] and recent Mamba-based moment retrieval models [7].

The BiMamba encoder consists of stacked bidirectional Mamba blocks. Each block has two temporal branches: a forward branch that processes the audio sequence in its original temporal order and a backward branch that processes the reversed sequence. The backward output is reversed again to recover the original order, concatenated with the forward output, and projected back to the model hidden dimension. A residual connection and layer normalization are then applied for stable optimization. This bidirectional structure allows each temporal frame to incorporate both preceding and following acoustic context while avoiding the quadratic cost of full self-attention over long audio sequences.

To further handle events with different temporal extents, we apply a lightweight multi-scale temporal fusion module after BiMamba encoding. The module combines the original high-resolution temporal representation with downsampled contextual representations and upsamples them back to the original temporal resolution. The resulting multi-scale representation is added through residual connections, enabling the model to capture both short local events and longer temporal patterns while preserving frame-level resolution for boundary prediction.

3.3. Quality-aware training and re-ranking

We retain the DETR-style Hungarian matching framework of the baseline and add a quality prediction branch to estimate the localization reliability of each predicted window. The model is trained with foreground/background classification, L1 span regression, temporal GIoU, saliency, and quality regression losses. For each matched prediction, the quality target is defined as the temporal IoU between the predicted window and the corresponding ground-truth moment. The IoU target is detached from the computation graph, allowing the quality branch to learn localization reliability without altering the span regression objective.

In audio moment retrieval, a candidate may be semantically relevant to the query while having inaccurate temporal boundaries. We therefore treat semantic relevance and localization reliability as complementary

ranking signals. During inference, we combine foreground probability p^{fg} with predicted quality $\hat{q}$ to obtain a quality-aware decoder score:

$$s_{\text{dec}} = p^{\text{fg}} \cdot \hat{q}. \quad (2)$$

Candidate windows are first ranked according to s_{dec} . For Feat.-Div. Ens.+LLM, CLAP-WavLM Ens.+LLM, and Dual-CLAP Ens.+LLM, we further apply a frozen audio-language LLM verifier to re-rank the top- K candidates, where $K = 10$. Feat.-Div. Ens. does not use LLM-based post-processing and relies only on decoder-based ensemble confidence.

For LLM verification, each candidate audio segment is cropped using its predicted start and end times and expanded on both sides by 50% of the candidate duration to provide additional acoustic context. The cropped audio, the text query, and the relative start and end positions of the candidate interval within the crop are given to Qwen2.5-Omni [8] as a frozen verifier. The verifier is prompted to answer whether the candidate interval accurately covers the moment described by the query. The normalized probability of the “Yes” answer, computed from the “Yes” and “No” logits, is used as the LLM relevance score s_{LLM} .

The final ranking score is computed as

$$s_{\text{final}} = (1 - \beta)s_{\text{dec}} + \beta s_{\text{LLM}}. \quad (3)$$

To diversify the final LLM-based variants, we evaluated interpolation weights $\beta \in \{0.30, 0.35, 0.45\}$ on the development splits using R1@0.7 as the primary criterion. We used $\beta = 0.35$, $\beta = 0.45$, and $\beta = 0.30$ for Feat.-Div. Ens.+LLM, CLAP-WavLM Ens.+LLM, and Dual-CLAP Ens.+LLM, respectively. The LLM verifier is used only as a post-processing re-ranker: it does not modify predicted temporal boundaries, update AMR model parameters, or use any ground-truth information from the hidden challenge evaluation set.

4. EXPERIMENTS

This section describes the experimental setup used to develop QAM-DETR. We first summarize the datasets and evaluation protocol, and then describe the feature extraction, model configuration, and training procedure. All model development is conducted using the public Clotho-Moment and CASTELLA splits. The hidden DCASE 2026 Task 6 evaluation set is used only for generating the final official submissions.

4.1. Datasets

We use Clotho-Moment for pretraining and CASTELLA for fine-tuning and system development. Clotho-Moment is a synthetic audio moment retrieval dataset constructed by overlaying short audio clips from Clotho onto long background recordings from Walking Tours [2]. It provides long audio recordings paired with text queries and temporal moment annotations, and is used only for pretraining.

CASTELLA is a human-annotated long-audio moment retrieval dataset with free-form captions and temporal boundaries [3]. It contains 1,009 training recordings, 213 validation recordings, and 640 test recordings. We use the official train, validation, and test splits without adding any manual annotations. After pretraining on Clotho-Moment, the model is fine-tuned on the CASTELLA training split. The CASTELLA validation split is used as the primary public split for model selection, while the CASTELLA test split is used as an additional public development benchmark to assess robustness of candidate systems. The hidden challenge evaluation set is not used for training, tuning, or model selection, and is used only for final submission.

4.2. Model setup

All audio and text features are extracted using pretrained models and precomputed before training. The feature extractors are kept frozen throughout pretraining, fine-tuning, validation, and inference. Audio features are extracted using a 1-second window with a 1-second hop, and no audio augmentation is applied. Each feature stream is projected into a shared hidden dimension inside QAM-DETR before gated feature integration.

LAION-CLAP [4] is used as the primary audio-language representation because it provides audio and text embeddings in a shared semantic space. To complement this representation, we additionally use MS-CLAP [5], WavLM [9], and RoBERTa [10]. MS-CLAP provides auxiliary CLAP-style audio-text features, WavLM provides fine-grained acoustic and temporal audio representations, and RoBERTa provides additional text-only semantic representations for query understanding. These auxiliary streams are integrated with the main LAION-CLAP representation through the learnable scalar gates described in Section 3.1.

Audio-language LLMs are not used for feature extraction, training, or boundary prediction. Qwen2.5-Omni [8] is used only as a frozen verifier for post-processing-based re-ranking in Feat.-Div. Ens.+LLM, CLAP-WavLM Ens.+LLM, and Dual-CLAP Ens.+LLM. It receives the predicted candidate audio crop and the corresponding text query, and re-scores the already predicted candidate windows without modifying their temporal boundaries or updating any model parameters.

4.3. Training

All models are trained in two stages. We first pretrain QAM-DETR on Clotho-Moment to learn general audio-text temporal grounding from synthetic moment annotations, and then fine-tune it on the CASTELLA training split for real long-audio moment retrieval. During both stages, pretrained feature extractors remain frozen and only the QAM-DETR modules are optimized.

We train the model using AdamW with precomputed audio and text features, without audio augmentation. The training objective follows the DETR-style Hungarian matching framework and consists of foreground/background classification, L1 span regression, temporal GIoU, saliency, and quality regression losses. The quality regression loss supervises the predicted localization quality using the temporal IoU of matched predictions, as described in Section 3.3.

4.4. Computational Efficiency

The QAM-DETR localization model contains 55.6 M trainable parameters. Since all pretrained audio and text feature extractors are frozen and their features are precomputed, only the QAM-DETR localization network is optimized during training. For a temporal sequence of length T , the bidirectional Mamba encoder requires two linear-time sequence passes and therefore scales linearly with T , whereas Transformer self-attention scales quadratically with temporal sequence length. The optional Qwen2.5-Omni verifier is not required for the base QAM-DETR inference and is applied only to the top- K candidates ($K=10$) as a post-processing step.

5. RESULTS

This section evaluates QAM-DETR from two perspectives. First, we conduct a cumulative ablation study on the CASTELLA validation and test splits to analyze the contribution of each architectural component. Second, we evaluate the final ensemble and re-ranking strategies on the CASTELLA development splits and the official DCASE 2026 Task 6 evaluation set. The ablation results focus on the non-ensemble model, while the final system results reflect the combined effect of feature diversity, model ensembling, and optional LLM-based re-ranking.

Table 1: Cumulative ablation results on the CASTELLA validation and test splits. Results are reported as mean $\pm$ standard deviation over five random seeds.

Configuration	val R1@0.5	val R1@0.7	val mAP	val mAP@0.5	val mAP@0.75	test R1@0.5	test R1@0.7	test mAP	test mAP@0.5	test mAP@0.75
DETR-style baseline with LAION-CLAP	39.32 $\pm$ 1.89	22.22 $\pm$ 2.11	17.89 $\pm$ 1.26	35.69 $\pm$ 1.37	15.80 $\pm$ 1.90	33.01 $\pm$ 1.64	16.42 $\pm$ 0.96	13.98 $\pm$ 0.81	28.94 $\pm$ 1.44	12.02 $\pm$ 0.69
+ BiMamba encoder	46.19 $\pm$ 0.66	29.20 $\pm$ 1.35	23.10 $\pm$ 0.58	42.11 $\pm$ 0.66	21.80 $\pm$ 0.80	38.66 $\pm$ 0.85	22.17 $\pm$ 0.74	18.32 $\pm$ 0.27	34.33 $\pm$ 1.03	16.96 $\pm$ 0.33
+ Quality prediction & ranking	50.46 $\pm$ 1.77	34.09 $\pm$ 1.60	26.33 $\pm$ 1.05	44.83 $\pm$ 1.53	25.28 $\pm$ 1.17	42.59 $\pm$ 1.22	25.32 $\pm$ 1.29	20.39 $\pm$ 1.14	36.84 $\pm$ 1.39	18.98 $\pm$ 1.48
+ Multi-scale temporal fusion	51.08 $\pm$ 2.18	34.60 $\pm$ 1.74	26.06 $\pm$ 1.21	45.87 $\pm$ 2.00	24.57 $\pm$ 1.04	44.05 $\pm$ 0.86	26.65 $\pm$ 0.81	21.03 $\pm$ 0.34	37.59 $\pm$ 0.54	19.92 $\pm$ 0.54
+ 3-way cross-attention	51.25 $\pm$ 2.38	34.43 $\pm$ 1.57	26.73 $\pm$ 1.00	45.73 $\pm$ 1.56	25.47 $\pm$ 1.15	45.26 $\pm$ 1.46	27.65 $\pm$ 0.77	21.82 $\pm$ 0.39	38.73 $\pm$ 0.73	20.54 $\pm$ 0.50
+ Multi-feature integration	52.33 $\pm$ 1.94	35.46 $\pm$ 2.01	28.11 $\pm$ 1.53	46.76 $\pm$ 1.46	27.08 $\pm$ 1.63	47.40 $\pm$ 1.51	29.00 $\pm$ 0.93	23.50 $\pm$ 0.76	40.74 $\pm$ 0.95	22.19 $\pm$ 0.88

5.1. Ablation study

We conduct a cumulative ablation study to quantify the contribution of each component in QAM-DETR. Starting from the DETR-style baseline with LAION-CLAP features, we progressively add the BiMamba encoder, quality regression loss, multi-scale temporal fusion, 3-way cross-attention, and multi-feature integration. Each row in Table 1 adds one component on top of the previous configuration. All results are reported as mean $\pm$ standard deviation over five random seeds.

Table 1 shows the largest gains from bidirectional temporal modeling and localization-quality estimation. BiMamba increases test R1@0.7 from 16.42% to 22.17%, and adding quality regression further raises it to 25.32%. These gains are consistent with the importance of surrounding temporal context and of distinguishing semantic foreground confidence from boundary accuracy.

Multi-scale temporal fusion and 3-way cross-attention provide moderate improvements in most test-side metrics, although they do not improve every validation metric. Multi-feature integration achieves the best overall performance among the ablated configurations, reaching 47.40% test R1@0.5, 29.00% test R1@0.7, and 23.50% test mAP. We use the full model as the base architecture for the final systems.

Because Table 1 follows a cumulative ablation protocol, each increment should not be interpreted as a fully isolated effect of that component. Within this protocol, the results support temporal reasoning and localization-aware ranking as central mechanisms in QAM-DETR’s query-conditioned localization pipeline, with multi-scale fusion, cross-modal interaction, and multi-feature integration providing complementary improvements.

5.2. Ensemble and re-ranking analysis

Based on the ablation results, the final systems use the full QAM-DETR architecture with feature-diverse ensembles and optional LLM-based re-ranking.

To increase candidate diversity, we use three feature combinations. Combination A uses LAION-CLAP, MS-CLAP, and WavLM-base-plus audio features with LAION-CLAP and MS-CLAP text features. Combination B uses LAION-CLAP and MS-CLAP audio-text features. Combination C uses LAION-CLAP together with WavLM-base, WavLM-base-plus, and WavLM-large audio features, and combines LAION-CLAP text features with RoBERTa. These combinations provide complementary information: CLAP-style features capture audio-text semantic alignment, WavLM features provide fine-grained acoustic and temporal cues, and RoBERTa provides additional textual semantic representations.

Feat.-Div. Ens. merges candidate predictions from feature combinations A, B, and C and ranks them using decoder-based confidence scores. Feat.-Div. Ens.+LLM uses the same candidate pool with LLM-based re-ranking. CLAP-WavLM Ens.+LLM and Dual-CLAP Ens.+LLM use ensembles based on feature combinations A and B, respectively, with LLM-based re-ranking. In all LLM-based variants, the verifier only re-scores top-ranked candidates and does not modify temporal boundaries.

Table 2 reports the performance of the four final systems on the CASTELLA development splits and the official DCASE 2026 Task 6

Table 2: Performance of the final system variants on the CASTELLA development splits and the official DCASE 2026 Task 6 evaluation set. All values are R1 scores (%).

System	Val R1@.5	Val R1@.7	Test R1@.5	Test R1@.7	Eval R1@.5	Eval R1@.7
Feat.-Div. Ens.	58.81	49.43	55.23	42.91	57.63	45.20
Feat.-Div. Ens.+LLM	64.77	53.12	59.02	45.14	62.15	48.02
CLAP-WavLM Ens.+LLM	67.05	50.00	58.87	45.14	63.84	48.59
Dual-CLAP Ens.+LLM	66.48	54.55	54.57	41.28	61.02	44.07

evaluation set. Unlike Table 1, which reports non-ensemble cumulative ablation results averaged over five random seeds, Table 2 reflects the combined effect of feature diversity, model ensembling, and optional LLM-based re-ranking. Therefore, the results in Table 2 should be interpreted as final system-level performance rather than the effect of a single architectural component.

On the official evaluation set, CLAP-WavLM Ens.+LLM achieves the best performance among the final system variants, with R1@0.5 of 63.84% and R1@0.7 of 48.59%, ranking joint first in DCASE 2026 Task 6. Importantly, Feat.-Div. Ens. and Feat.-Div. Ens.+LLM use the same ensemble candidate pool and differ only in LLM-based re-ranking. This controlled comparison improves evaluation R1@0.7 from 45.20% to 48.02%, providing direct evidence that the verifier can improve candidate ranking within this configuration. However, Dual-CLAP Ens.+LLM obtains lower R1@0.7 than Feat.-Div. Ens.+LLM and CLAP-WavLM Ens.+LLM, suggesting that the verifier is most effective when accurate temporal proposals are already included among the top-ranked candidates. Since the LLM verifier does not refine temporal boundaries, its benefit is limited by the localization quality of the decoder-generated windows.

6. CONCLUSION

We presented QAM-DETR, a task-oriented quality-aware DETR framework for long-audio moment retrieval. Its localization pipeline connects selective integration of heterogeneous pretrained audio-text representations, query-conditioned bidirectional temporal reasoning, and localization-quality-aware candidate ranking. Gating and progressive 3-way cross-modal interaction construct a query-conditioned temporal representation; BiMamba with multi-scale fusion models long-range context across temporal resolutions while preserving frame-level boundary resolution. The quality branch complements semantic foreground confidence with predicted localization reliability. Five-seed cumulative ablations show the largest gains from bidirectional temporal modeling and quality-aware prediction, with the remaining components providing complementary improvements.

For the final system variants, we further investigated feature-diverse ensembling and frozen audio-language LLM-based candidate re-ranking. On the official DCASE 2026 Task 6 evaluation set, our best system variant achieved R1@0.5 of 63.84% and R1@0.7 of 48.59%, ranking joint first in the challenge. These results suggest that long-audio moment retrieval requires not only semantic audio-text matching, but also reliable temporal context modeling and localization-aware candidate ranking. The mixed effect of LLM-based re-ranking further indicates that semantic verification is most effective when accurate temporal proposals are already included in the candidate pool. Future work includes boundary refinement, stable feature fusion, and efficient verification for long audio.

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